MUHAMMAD ZAYAB ANSARI

AI / ML Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate specialising in Artificial Intelligence and Machine Learning. Experienced in architecting RAG pipelines and production-grade REST APIs using FastAPI, LangChain, and LLMs, with deployment on Hugging Face Spaces and cloud-compatible Docker/Nginx stacks. Proven track record delivering end-to-end AI systems across NLP, Computer Vision, and Generative AI — from data preprocessing and model training to containerised deployment. Familiar with MLOps best practices including CI/CD via GitHub Actions, model versioning, and experiment tracking. Seeking a junior AI/ML Engineer role to contribute to impactful, innovative products.

EXPERIENCE

AI Engineering Intern | SMIT Bootcamp 5.0 | *Jan 2025 – Present*

- Developed and deployed AI/ML models for real-world use cases including NLP, Computer Vision, and Generative AI applications; models deployed via Docker containers on Hugging Face Spaces and cloud-compatible environments.
- Built end-to-end ML pipelines covering data preprocessing, feature engineering, model training, evaluation, hyperparameter tuning, and deployment; tracked experiments using structured logging and versioned model artefacts.
- Collaborated on team projects integrating LLM-based solutions with FastAPI backends and Streamlit frontends, following REST API versioning and CI/CD best practices using GitHub Actions.
- Worked with tools including Python, PyTorch, TensorFlow, Scikit-Learn, LangChain, FAISS, Docker, Hugging Face Transformers, and cloud-hosted inference endpoints.

Freelance AI / ML Engineer | Fiverr (Self-Employed) | *Mar 2024 – Present*

- Delivered 15+ freelance ML, NLP/RAG, and Computer Vision projects for international clients across industries including healthcare, legal, and e-commerce.
- Built custom AI solutions including chatbots, image classifiers, data analysis pipelines, and Generative AI integrations; deployed solutions via Docker on client-managed cloud infrastructure (AWS-compatible, GCP-compatible).
- Maintained 5-star client satisfaction through full project lifecycle ownership: requirements gathering, model development, A/B testing, iterative improvement, and client delivery.

EDUCATION

B.E. Software Engineering | Sir Syed University of Engineering & Technology (SSUET), Karachi | *2022 – 2026*

- Studied core software development, algorithms, system design, databases, and object-oriented programming.
- Final Year Project: Pakistan LegalAI Platform — a production-ready RAG-based legal assistance system with CI/CD pipeline and containerised cloud deployment.
- Relevant coursework: Big Data Analytics (Hadoop, Spark, MapReduce), OOP, Database Systems, Computer Networks, Machine Learning Fundamentals.

Data Science & AI Bootcamp | SMIT — Saylani Mass IT Training | *2025 – 2026*

- Supervised/unsupervised Machine Learning, Deep Learning (CNNs, Transfer Learning, Transformer architectures), and Generative AI.
- Hands-on training in LLMs, RAG pipelines, prompt engineering, RLHF concepts, agentic AI workflows using LangChain, and MLOps fundamentals including experiment tracking and model monitoring.

PROJECTS

AI Digital Ghost — Personality Preservation System

- Production-ready full-stack AI system that digitally preserves a person's personality, communication style, voice, and memories by ingesting WhatsApp chats, emails, voice recordings, and photos — enabling authorised users to interact with a personalised AI ghost in the deceased or absent person's exact tone and style.
- AI core: LLaMA 3 (8B) via Ollama for local, private LLM inference; Sentence Transformers + ChromaDB for semantic memory search and retrieval-augmented generation; style transfer pipeline extracting catchphrases, humor patterns, and emotional tone from raw conversation data.
- Voice pipeline: OpenAI Whisper for speech-to-text transcription of voice notes; Coqui TTS for voice cloning, enabling responses in the original speaker's synthesised voice.
- Full-stack architecture: Next.js + TypeScript + Tailwind CSS frontend; Python FastAPI backend with JWT authentication and role-based access control; PostgreSQL for user data; ChromaDB for memory vectors; Redis for response caching; fully containerised with Docker Compose and deployed on AWS/GCP.
- Privacy-first design: all LLM inference runs locally via Ollama (no data leaves the server); access restricted to authorised users only via JWT; ethical consent framework built into onboarding flow.

Pakistan LegalAI Platform — Final Year Project

- Production-ready RAG-powered legal AI platform ingesting 12+ Pakistani laws into a FAISS vector index; achieves 95%+ retrieval relevance on legal queries using LLaMA 3-70B (Groq API) with average response latency under 3 seconds.
- Features: Urdu/English bilingual chatbot, court case search across 21 real judgments (PLD/SCMR citations), legal PDF generation, Know Your Rights module, and JWT-authenticated 25+ REST API endpoints; deployed via Docker + Nginx on a cloud-compatible Linux server.
- MLOps: containerised with Docker Compose, CI/CD via GitHub Actions, environment-based config management, and structured API versioning.
- Tech: Python, FastAPI, React 18, LangChain, FAISS, LLaMA 3-70B, Groq API, Sentence Transformers, PostgreSQL, Docker, Nginx, ReportLab, Tailwind CSS.

ClinIQ — Clinical AI Platform

- Microservices-based clinical AI platform combining Random Forest, CNN/ONNX, and LLaMA 3.3-70B for multi-modal medical diagnosis assistance; deployed on Hugging Face Spaces (cloud).
- Integrated 3 independent ML models into a unified FastAPI microservices backend with Streamlit frontend, reducing diagnostic query response time to under 2 seconds; each service independently containerised for scalable cloud deployment.
- Implemented model monitoring hooks and structured logging for inference latency and prediction confidence tracking.

Real-Time Brain Tumor Detection System

- YOLOv11-based brain tumor detection system achieving 93.5% mAP accuracy at 30 FPS inference on medical MRI image streams; model trained with data augmentation and transfer learning from a pretrained YOLO backbone.
- Built interactive Streamlit web interface for real-time model visualisation, annotation display, and user upload; deployed on Hugging Face Spaces for public demo.

Real-Time License Plate Detection System

- YOLOv11-based license plate detection system supporting both image and video inputs via an intuitive Gradio web interface; handles .jpg, .png, .mp4, .avi, .mov, and .mkv formats with automatic format detection.
- Detects and annotates plates with bounding boxes and confidence scores in real time; video processing tracks total detection count across all frames at native FPS with OpenCV VideoWriter output.
- Tech: Python, YOLOv11 (Ultralytics), OpenCV, Gradio, NumPy — deployed as a containerised end-to-end CV application.

Flight Delay Prediction System

- End-to-end ML pipeline predicting flight delays from historical aviation data; achieved 89% precision/recall balance using XGBoost with extensive temporal and categorical feature engineering.

- Addressed class imbalance via SMOTE; benchmarked and A/B tested 5 classification algorithms using cross-validation to select the best-performing model.

Full-Stack Custom Teachable Machine

- Google Teachable Machine clone using MobileNetV3 (PyTorch) for feature extraction and Scikit-Learn for classification; supports real-time custom model training in the browser with under 5-second train time for 3-class problems.
- FastAPI backend + Streamlit frontend; packaged as a portfolio-grade open-source project on GitHub with Docker support.

AI Chat & Image Generation Web App

- Web-based AI assistant with dual chat and image generation modes using Groq API, Hugging Face Inference API, and Pollinations.ai; handles real-time streaming responses, loading states, error fallback, and mode switching in vanilla JavaScript.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS

AI / ML: Scikit-Learn, TensorFlow, PyTorch, XGBoost, LightGBM, Random Forest, CNNs, Transfer Learning (VGG16, MobileNetV3), SMOTE, Transformers

Generative AI: LLMs, Prompt Engineering, RLHF (concepts), OpenAI API, Groq API, Hugging Face, Model Fine-Tuning, RAG, Local LLM inference (Ollama)

Agentic AI: LangChain, LangGraph, RAG Pipelines, FAISS, ChromaDB, Sentence Transformers, Vector Databases

Computer Vision: YOLOv11, ONNX, OpenCV, Image Preprocessing, Data Augmentation, Feature Extraction, MRI Analysis

Voice & Speech: Whisper (STT), Coqui TTS, Voice Cloning, Audio Preprocessing

Backend & APIs: FastAPI, REST API Design & Versioning, SQLAlchemy, PostgreSQL, SQLite, Redis, JWT Auth, Docker, Docker Compose, Nginx, Microservices

Frontend: Next.js, React 18, TypeScript, Tailwind CSS, Streamlit, Gradio

Cloud & MLOps: AWS/GCP deployment, Hugging Face Spaces, CI/CD (GitHub Actions), Model Versioning, Experiment Tracking, Model Monitoring

Data & Analytics: EDA, Feature Engineering, A/B Testing, Predictive Modeling, Data Visualization (Matplotlib, Seaborn), Pandas, NumPy

Tools: Git/GitHub, Jupyter Notebook, VS Code, Linux/Bash, Redis

CERTIFICATIONS & ACHIEVEMENTS

Cisco Networking Academy

- Python Essentials 1 — Dec 2025
- Python Essentials 2 — 2026

Kaggle

- Computer Vision • Intro to Deep Learning • Intro to Machine Learning • Data Cleaning • Data Visualization • Python

Saylani Mass Training Program

- Coding Night 2026 — Hackathon Participant